

From Regional Dialects to Standard Written Marathi: A Neural Sequence-to-Sequence Normalization Framework

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Abstract

Regional sub-dialects of Indic languages suffer from acute resource scarcity and an absence of aligned parallel corpora for generative text normalization. In this work, we present an end-to-end neural sequence-to-sequence framework for normalizing non-standard spoken transcripts from three regional Marathi varieties—Southern Konkan, Northern Konkan, and Varhadi—into formal Standard Written Marathi. To address severe data constraints, we introduce an automated synthetic parallel expansion pipeline pairing 1-call-per-sentence Gemma-4 generation with a multi-tier verification engine comprising Devanagari Unicode rule filtering and LLaMA-3.1-8B semantic auditing, expanding the parallel dataset from 16,163 to 32,335 verified pairs. We systematically evaluate IndicBART (244M) and mT5-small (300M) across single-dialect and multi-dialect configurations under stratified 5-fold cross-validation and the held-out RESPIN test benchmark (2,170 utterances). Experimental findings demonstrate that mT5-small achieves 69.51 BLEU and 84.58 chrF++ on the expanded multi-dialect benchmark, outperforming IndicBART by 12.39 BLEU. Crucially, multi-tier verification prevents synthetic quality collapse (+14.03 BLEU recovery over unverified data), establishing verified synthetic augmentation as an effective foundation for low-resource dialect normalization.

Keywords: Dialect Normalization, Low-Resource NLP, Marathi Dialects, Sequence-to-Sequence Modelling, Synthetic Data Augmentation, mT5, IndicBART.

1 Introduction

Rapid development of deep learning architectures and self-attention mechanisms (Vaswani et al., 2017) has made it possible to develop systems capable of understanding and generating natural language. Open-weight models have gained popularity to process text in various domains as they provide a finer control over system architecture, runtime environment, and training data (Dubey et al., 2024). Yet current language systems tend to work efficiently with standard high-resource languages. Queries based on non-standard input yield unexpected and undesired outputs (Joshi et al., 2020). Speakers of regional dialects face many issues in systems that use these models across automated speech recognition, machine translation and other language processing systems (Koenecke et al., 2020; Zampieri et al., 2020).

Dialect normalization is the task of translating non-standard regional dialects into standard written language. The main challenge to be addressed while designing such systems is the preservation of the meaning of original text (Blaschke et al., 2024). It serves to be an essential layer for digital linguistic inclusion. Countries having large and diverse speech communities, such as India, which has 22 official languages and hundreds of regional dialects spoken by 1.4 billion people, regional varieties have remained unattended in computational linguistic research. An example of such linguistic disparity is Marathi, an Indo-Aryan language spoken by over 83 million people in western India (Kulkarni-Joshi, 2023). Development of reliable pipelines for regional Indic dialects have faced these fundamental challenges:

1. *Resource Scarcity*: Parallel corpora mapping the regional dialects to standard written

language do not exist. Curation of such sets by experts fluent in multiple dialects is both expensive and unscalable (Blaschke et al., 2024; Dhasmana et al., 2026).

2. *Morphological and Dialectal Heterogeneity*: Marathi exhibits rich morphology where different words undergo varied complex transformations across different regions where the dialects are spoken (Ansary et al., 2024). Consequently, evaluations vary significantly across individual dialects—ranging from systematic suffix alternation in Varhadi to intense lexical and phonetic divergence in Northern and Southern Konkan (Joshi et al., 2025).
3. *Underperformance of Multilingual Models*: Performance of pre-trained models like IndicBART (Dabre et al., 2022), IndicNLP Suite (Kakwani et al., 2020), and mT5 (Xue et al., 2021) seems to drop drastically on non-standard language variants due to fragmentation during tokenisation and out-of-vocabulary phonological shifts (Joshi et al., 2020; Kumar et al., 2024).

Our work mainly tries to investigate and conclude (a) how do normalisation scores vary across regional Marathi dialects which have different degrees of variations as compared to standard Marathi; (b) how effectively can existing sequence-to-sequence architectures adapt to dialect-to-standard mapping under task-specific conditioning; and (c) whether automated synthetic data expansion can accurately solve the problem of data scarcity and eliminate cross-dialect hallucinations and grammatical errors (de Gibert et al., 2024; Ji et al., 2023). To address these questions, we construct an end-to-end normalization framework combining LLM-based synthetic generation, multi-tier syntactic and semantic verification, and conditioned transformer fine-tuning across both single-dialect and multi-dialect settings.

Our core contributions are:

- *Parallel Corpus Generation*: Based on transcripts from the RESPIN dataset (Kumar et al., 2025), we construct a parallel corpus for three of the Marathi dialects (Southern Konkan, Northern Konkan, and Varhadi) mapped to Standard Marathi.
- *Automated Synthetic Augmentation with Verification*: We design a data expansion pipeline paired with a multi-tier verification engine, which has rule-based script filtration and LLM-based semantic inspection.

2 Related Work

2.1 Regional Dialect Standardization across Language Families

Early approaches in the context of European languages used Character-Level Statistical Machine Translation (CSMT) and phrase-based models for Swiss German Dialects (Scherer and Ljubešić, 2016) which were later replaced by pre-trained language models (Lusetti et al., 2018; Kresic and Abbas, 2024). Similar approaches have been developed for Finnish, Swedish, and Estonian dialects (Partanen et al., 2019; Hämäläinen et al., 2020, 2022). A study of four language families by Kuparinen et al. (Kuparinen et al., 2023) suggested that fine-tuned sequence-to-sequence transformers surpassed the rule-based and SMT baselines. Other than European countries, East Asian and African-Asian varieties have also shown similar patterns. Japanese regional dialects have been processed using phrase-level LSTM modules (Abe et al., 2018), while Mandarin scripts relied on acoustic-phonetic mapping (Zhao and Chodroff, 2022). The Conventional Orthography for Dialectal Arabic (CODA) framework (Habash et al., 2012) in the landscape of Arabic languages gave guidelines for Modern Standard Arabic (MSA) which led to multi-city translation benchmarks (MADAR (Bouamor et al., 2018), NADI (Abdul-Mageed et al., 2023)) and morphology-oriented prompt design for LLMs (Eryani et al., 2020; Dimakis et al., 2025).

2.2 Indic Dialect Landscape and Marathi NLP

Even though pre-trained models such as IndicBART (Dabre et al., 2022) and mT5 (Xue et al., 2021), which are trained on multilingual corpora cover major languages, they lack under-

standing of regional sub-dialects. Models based on a single language like L3Cube-MahaNLP (Joshi et al., 2022) offer pre-trained Marathi models for classification, but do not support generative text standardization. INDIC-DIALECT (Kumar et al., 2024) considers Hindi varieties (Bhojpuri, Magahi, Maithili), for evaluation via WFST frameworks for text normalization while preserving the underlying semantics (Pratap et al., 2024). However, Marathi dialects were primarily dealt for acoustic/ASR classification (Mulla and Kumar, 2022; Pawar et al., 2026), or zero-shot LLM retrieval (Gaikwad et al., 2025; Vyavhare et al., 2026). Our work fills this gap by presenting an end-to-end neural fine-tuning pipeline with evaluations and data augmentation strategies.

2.3 LLM Distillation and Multi-Tier Verification

LLMs can be used to suffice for data scarcity in various tasks (Wang et al., 2022; Taori et al., 2023; Mukherjee et al., 2023). Knowledge distillation strategies can be employed to train compact and efficient models based on the outputs generated by a larger teacher model (Hsieh et al., 2023; Pecher et al., 2024; de Gibert et al., 2024). But the use of such techniques for low-resource regional dialects involves significant amount of hallucinations and noise (de Gibert et al., 2024; Ji et al., 2023). We solve this by utilising Gemma-4 (Google DeepMind Gemma Team, 2026) with a robust morphologically aware system prompt and a multi-tier verification system that ensures integrity of the data generated.

3 Methodology

3.1 Original Corpus Curation and Foundation in RESPIN

Base sentences for our task are drawn from the domain-specific transcripts of the RESPIN-S1.0 initiative (Kumar et al., 2025), developed at IISc Bangalore under the Gates Foundation project. While RESPIN has audio recordings and their transcripts across agriculture and banking domains for Indian languages and their dialects, it lacks aligned parallel dialect-to-target language translations required for efficient normalisation. The large speaker diversity ensures broad coverage of intra-dialect phonetic and lexical variation, while the balanced gender and domain splits reduce systematic bias in our derived parallel corpus. Detailed corpus statistics and demographic splits are summarized in Table 1 and Figure 1.

Table 1: RESPIN-S1.0 Marathi Source Corpus Statistics

Code	Dialect Name	Utterances (%)	Duration (h)	Speakers	Prompts
D1	Southern Konkan (Malvani)	203,651 (25.14%)	247.31	732	5,838
D2	Northern Konkan (Ahirani)	198,560 (24.52%)	265.25	557	5,825
D3	Standard Marathi (Puneri)	208,850 (25.79%)	255.79	608	6,077
D4	Varhadi (Vidarbha)	198,873 (24.55%)	257.70	747	5,330
Total	All Dialects	809,934 (100%)	1,026.06	2,644	23,069

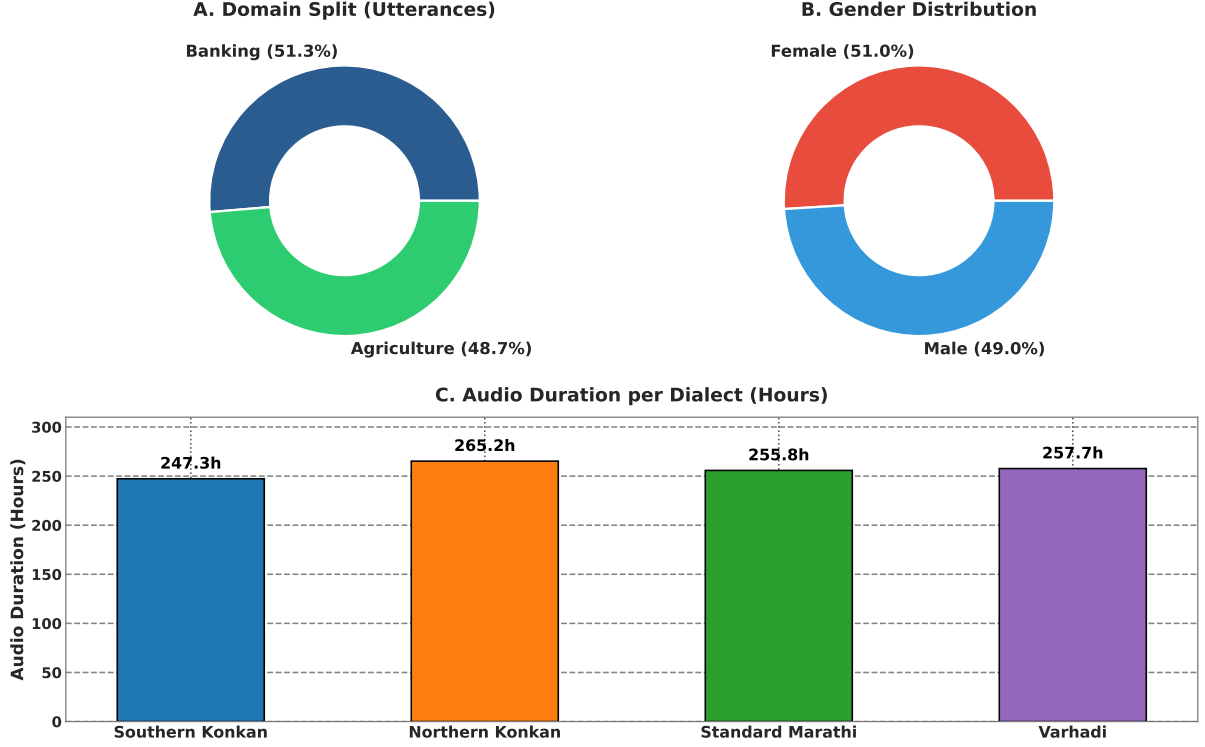


Figure 1: RESPIN-S1.0 Marathi source corpus demographics: (A) domain distribution; (B) gender split; (C) audio recording duration per dialect variant.

3.2 Target Dialects and Linguistic Properties

We target the following major Marathi dialects that span the primary regions of Maharashtra and exhibit distinct linguistic and phonetic patterns based on the RESPIN corpus:

1. **Southern Konkan (Konkan Coast)**: Characterized by shortening of final-word vowels, softening of dental consonants, and different auxiliary verb forms. Lexical differences include systematic vowel shifts: standard जातो (“he goes”) becomes Southern Konkan जातां; standard कुठे (“where”) becomes कुठंय. Domain-specific terminology generally remains untouched but often accompanied by Konkani words.
2. **Northern Konkan (Khandesh Region)**: Heavily influenced by neighbouring Gujarati and Bhili language families, thus it exhibits significant consonant simplification and dental consonant shifts. Notable transformations include: standard कशाला (“why”) → Northern Konkan कसाले; standard मुलगा (“boy”) → Northern Konkan पोर. Northern Konkan possesses the highest degree of lexical divergence among the three target dialects.
3. **Varhadi (Vidarbha/Eastern Maharashtra)**: Spoken in Vidarbha, it exhibits characteristic differences and verb suffix modifications. Standard interrogative का? (“why?”) becomes Varhadi काऊन?; verbal forms undergo regular suffix mapping patterns. Among the three dialects, Varhadi maintains closest structural similarity to Standard Marathi as transformations are more systematic.

3.3 Synthetic Parallel Data Generation Pipeline

To deal with data scarcity, we developed an automated synthetic data translation and generation pipeline powered by Gemma-4 (Google DeepMind Gemma Team, 2026) with 12B parameters configuration, accessed via Ollama. The complete end-to-end normalization architecture is illustrated in Figure 2. We considered several design decisions to ensure generation quality to be maximised:

1. **1-Call-Per-Sentence Generation:** We observed batch generation caused hallucination and sentence alignment errors across dialect boundaries. Therefore, we used a 1-call-per-sentence translation regime.
2. **Preservation of Meaning and Domain Constraints:** System prompts strictly instructed the model that while colloquial grammatical markers and phonetic spellings must be transformed into standard written Marathi, the standard output must preserve exact meaning, domain terminology (agriculture/banking), and named entities without creative embellishment.

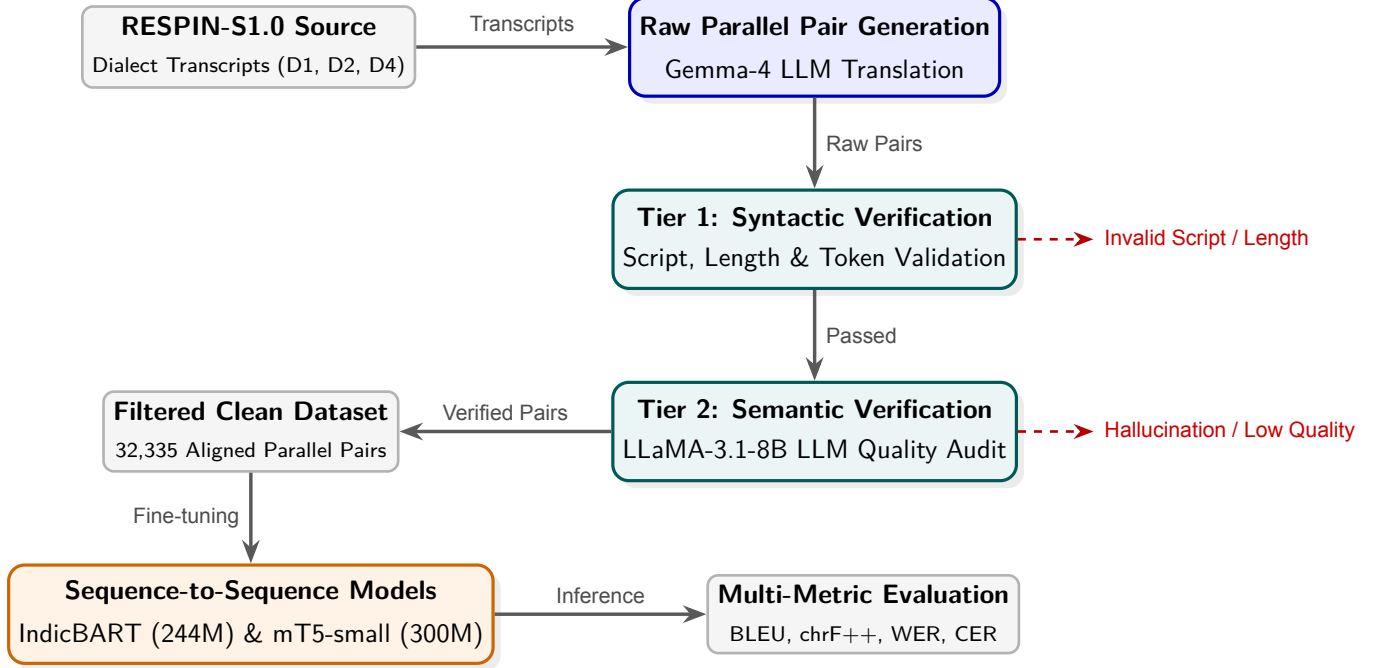


Figure 2: End-to-end multi-dialect Marathi text normalization framework: raw pair generation using Gemma-4, multi-tier syntactic/semantic verification, and conditioned sequence-to-sequence model fine-tuning.

3.4 Verification and Quality Filtering Engine

LLMs can generate hallucinated texts, script corruption, or flawed target sequences when dealing with regional sub-dialects. We implemented a two-tier verification engine to purge flawed sentence pairs before model training:

1. **Tier 1: Rule-Based Syntactic Filter.** Automated validation checks implement Indic Abugida script normalization principles (Ansary et al., 2024), including:
 - **Script Consistency:** Discards samples containing non-Devanagari characters (e.g., Latin script, Chinese glyphs, stray punctuation artifacts).
 - **Length Ratio Constraint:** Discards pairs where sentence length ratio $\frac{\text{len}(\text{Target})}{\text{len}(\text{Source})} \notin [0.5, 2.0]$.
 - **Minimum Token Count:** Both source and target must contain ≥ 3 whitespace-separated tokens.
2. **Tier 2: LLM Semantic Verification.** A secondary LLM evaluator using llama-3.1-8b-instant (Dubey et al., 2024) grades translation accuracy on a 1–5 semantic scale under strict prompt guidelines. Pairs with semantic scores < 3 are purged.

Through this multi-tier verification process, 3,751 low-quality sentence pairs were purged from the unverified corpus (19,914 pairs), preventing error propagation into downstream models. The dataset composition across original, synthetic, and final clean partitions is illustrated

in Figure 3.

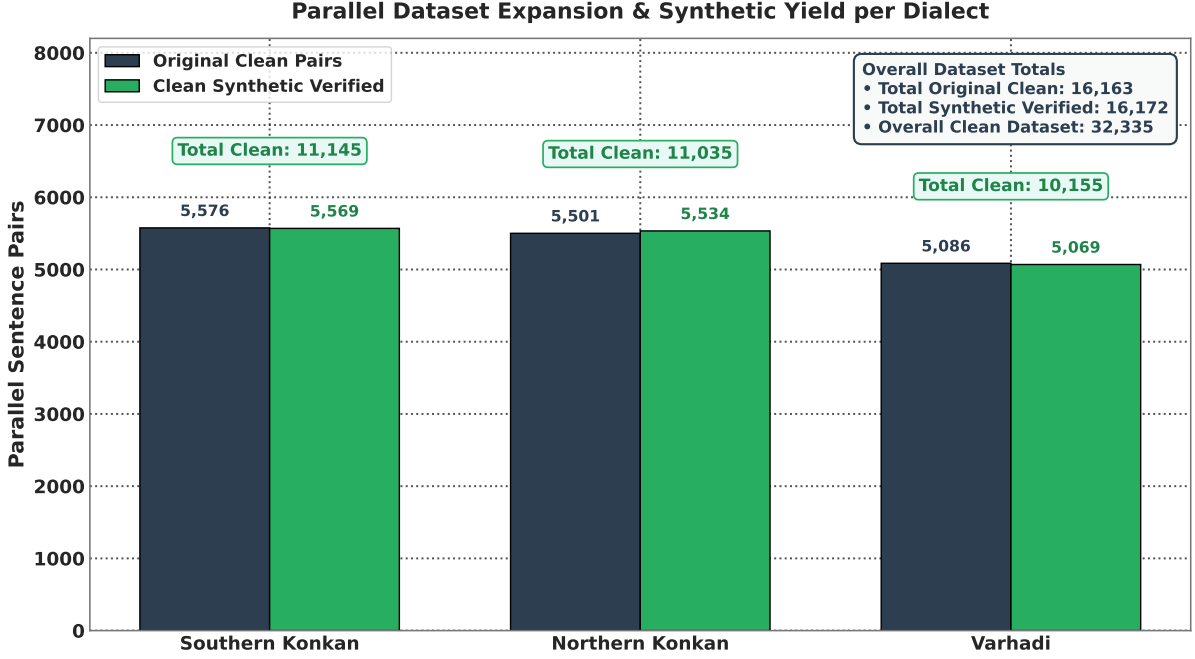


Figure 3: Parallel dataset expansion breakdown per Marathi dialect, showing original clean pairs, synthetic verified pairs, and the final expanded corpus totals.

3.5 Model Fine-Tuning and Conditioning

We benchmark two pre-trained sequence-to-sequence architectures across both single-dialect and multi-dialect configurations: IndicBART (244M) (Dabre et al., 2022) and mT5-small (300M) (Xue et al., 2021).

To enable unified multi-dialect training where a single model normalizes all regional variants, we implement special token conditioning prefixes:

- **Task Normalization Prefixes:** A normalisation tag (`<normalize>`) is attached that instructs the encoder to perform the dialect-to-standard text transformation.
- **Dialect-Source Identification Tags:** Source sentences are prefixed with dialect markers, `<D1>` for Southern Konkan, `<D2>` for Northern Konkan, and `<D4>` for Varhadi for explicitly conditioning the model particularly in the multi-dialect training scenario.
- **Target Standardization Tags:** Decoder target conditioning is done using standard language tags (`<standard_mr>`), enforcing the output alignment into formal Standard Written Marathi.

Table 2 details the underlying architectural properties and hyperparameter configurations for IndicBART (244M) and mT5-small (300M).

Table 2: Detailed Architecture Comparison: IndicBART (244M) vs. mT5-small (300M)

Property	IndicBART (244M)	mT5-small (300M)
Base Architecture	mBART-50 (BART Enc-Dec)	T5 Encoder-Decoder
Position Encoding	Absolute sinusoidal	Relative positional bias
Vocabulary Size	64,000 SentencePiece	250,000 SentencePiece (mC4)
Pre-training Data	11 Indic + English	101 languages (mC4)
Script Approach	Devanagari-unified	Native multi-script
Target Conditioning	Required <code><2mr></code> token	Raw text-to-text (no prefix)
Learning Rate	5×10^{-5} (linear warmup)	5×10^{-4} (AdaFactor)

3.6 Evaluation Protocol

We report model performance across two distinct evaluation settings:

1. **Mean 5-Fold Cross-Validation:** To evaluate model stability and in-domain generalization across the curated parallel datasets, models are trained and evaluated under an 85/15 stratified 5-fold cross-validation scheme (reported in Section 4.1–4.4, Tables 3–9, and Figures 4–8).
2. **Held-Out RESPIN Test Benchmark:** To measure true zero-shot/unseen generalization, models trained on the complete dataset are evaluated on the official held-out benchmark set (IISc_RESPIN_test_mr), consisting of 2,170 unseen utterances across all target sub-dialects (reported in Section 4.2, Table 7, and Figure 6).

Results are reported using metrics that are evaluated using SacreBLEU (Post, 2018) and Word Error Rate (WER) scripts:

- **BLEU** (Bilingual Evaluation Understudy): BLEU-4 with 4-gram precision, brevity penalty, with an exponential smoothing method.
- **chrF++**: Character n-gram F-score combined with word unigrams and bigrams.
- **Normalized WER / CER**: Word Error Rate and Character Error Rate after Devanagari Unicode script standardization.

4 Findings

To ensure transparent analysis, our findings are reported across two distinct evaluation protocols: (1) **Mean 5-Fold Cross-Validation** (Section 4.1, Tables 3–6, Figures 4–5): Evaluates in-domain model stability and dialectwise baseline performance across 5 stratified train/test partitions; and (2) **Held-Out RESPIN Test Benchmark & Ablation Evaluations** (Sections 4.2–4.4, Tables 7–9, Figures 6–8): Evaluates true unseen generalization on the official held-out test dataset comprising 2,170 utterances (IISc_RESPIN_test_mr) after training models on the complete dataset (single test run, without cross-validation).

4.1 Single-Dialect vs. Multi-Dialect 5-Fold Cross-Validation Results

Cross-validation performance on original single-dialect datasets is reported in Table 3.

Table 3: Benchmark Performance Matrix on Original Datasets: **Single-Dialect Models** (5-Fold CV)

Dialect Name	IndicBART (244M)		mT5-Small (300M)	
	BLEU	chrF++	BLEU	chrF++
Southern Konkani (D1)	48.54	78.35	46.31	74.43
Northern Konkani (D2)	60.58	80.30	60.31	77.34
Varhadi (D4)	76.63	90.93	81.00	91.21

Inference from Table 3 (Morphological Regularity vs. Lexical Divergence): Baseline performance highlights the structural disparity across Marathi dialects. Both models achieve their highest scores on Varhadi (76.63–81.00 BLEU and 90.93–91.21 chrF++), whereas coastal varieties (Southern and Northern Konkani) register lower baseline scores (46.31–60.58 BLEU). In real-world deployment, regional varieties with systematic grammatical and suffix alternations (like Vidarbha Marathi) are far easier for neural seq2seq models to normalize reliably with minimal supervision than coastal dialects that incorporate extensive vocabulary divergence, phonological shifts, and contact-language loanwords.

Table 4 presents the dialectwise evaluation breakdown when training a single unified multi-dialect model on the combined original data.

Table 4: Benchmark Performance Matrix on Original Datasets: **Multi-Dialect Dialect-wise Breakdown** (5-Fold CV)

Evaluated Subset	IndicBART (244M)		mT5-Small (300M)	
	BLEU	chrF++	BLEU	chrF++
Southern Konkani (D1)	26.21	53.15	48.28	75.04
Northern Konkani (D2)	47.59	67.16	62.35	78.92
Varhadi (D4)	72.75	85.63	79.57	90.51

Inference from Table 4 (Cross-Dialect Parameter Interference): Pooling all dialects into a unified multi-dialect model reveals significant capacity and interference effects: IndicBART suffers catastrophic cross-dialect interference on Southern Konkani (dropping by 22.33 BLEU to 26.21 BLEU and 53.15 chrF++), whereas mT5-small’s 250k vocabulary and relative positional bias mitigate cross-dialect interference, maintaining 48.28 BLEU on Southern Konkani and achieving an overall multi-dialect average of 63.29 BLEU and 81.37 chrF++.

Table 5 evaluates single-dialect models trained on synthetically expanded corpora.

Table 5: Benchmark Performance Matrix on Synthetically Expanded Datasets: **Single-Dialect Models** (5-Fold CV)

Dialect Name	IndicBART (244M)		mT5-Small (300M)	
	BLEU	chrF++	BLEU	chrF++
Southern Konkani (D1)	47.25	64.69	65.10	82.88
Northern Konkani (D2)	40.51	62.21	62.07	79.29
Varhadi (D4)	73.62	86.75	78.89	90.59

Inference from Table 5 (Scaling Dynamics in Isolated Dialects): Doubling single-dialect corpora via verified synthetic data reveals contrasting scaling dynamics: IndicBART experiences slight capacity saturation across isolated dialects (47.25 BLEU on Southern Konkani and 40.51 BLEU on Northern Konkani), while mT5-small leverages the synthetic pairs effectively,

surging to 65.10 BLEU (+18.79) on Southern Konkani and 62.07 BLEU on Northern Konkani with chrF++ exceeding 79.29–90.59.

Table 6 presents the dialectwise evaluation of the unified multi-dialect model trained on the complete synthetically expanded corpus (32,335 pairs).

Table 6: Benchmark Performance Matrix on Synthetically Expanded Datasets: *Multi-Dialect* Dialectwise Breakdown (5-Fold CV)

Evaluated Subset	IndicBART (244M)		mT5-Small (300M)	
	BLEU	chrF++	BLEU	chrF++
Southern Konkani (D1)	52.06	72.15	66.62	83.57
Northern Konkani (D2)	49.08	69.90	62.72	79.63
Varhadi (D4)	70.27	84.27	78.73	90.46

Inference from Table 6 (Synthetic Augmentation & Gradient Conflict Resolution): Evaluating the unified multi-dialect models on 32,335 verified pairs demonstrates the full benefit of synthetic data expansion. IndicBART completely recovers from cross-dialect interference to reach 52.06 BLEU (+25.85 gain over original multi-dialect) on Southern Konkani and 57.12 overall BLEU. Concurrently, mT5-small reaches peak performance across all subsets (66.62 BLEU on Southern Konkani, 62.72 BLEU on Northern Konkani, and an overall combined benchmark of 69.51 BLEU and 84.58 chrF++), confirming that synthetic scaling resolves gradient conflicts in multi-dialect seq2seq architectures.

Comparative single-dialect 5-fold cross-validation performance shifts are illustrated in Figure 4.

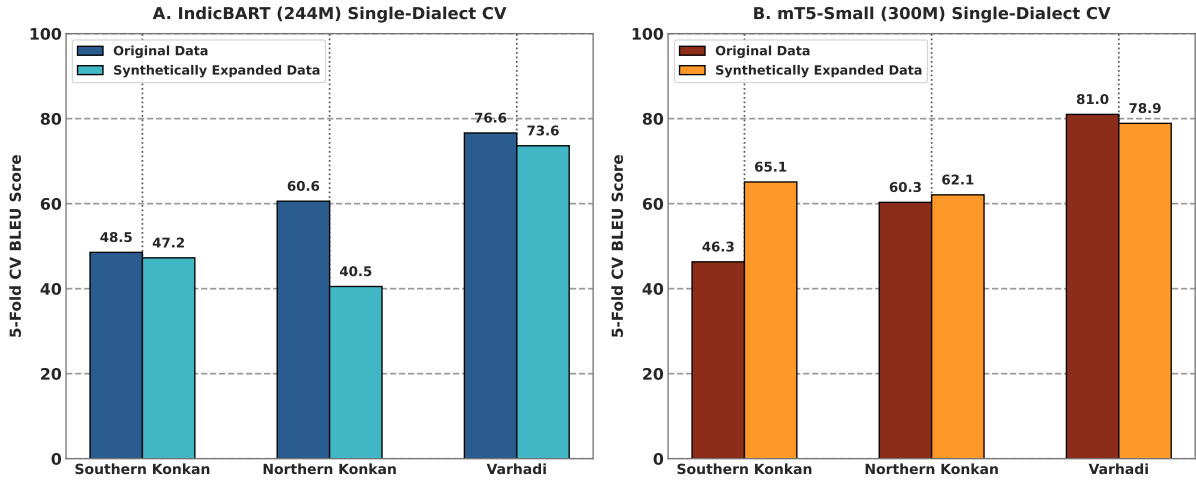


Figure 4: Single-dialect 5-fold cross-validation performance shift comparing Original and Synthetically Expanded models for IndicBART (244M) and mT5-small (300M).

The multi-dialect performance matrix is shown in Figure 5.

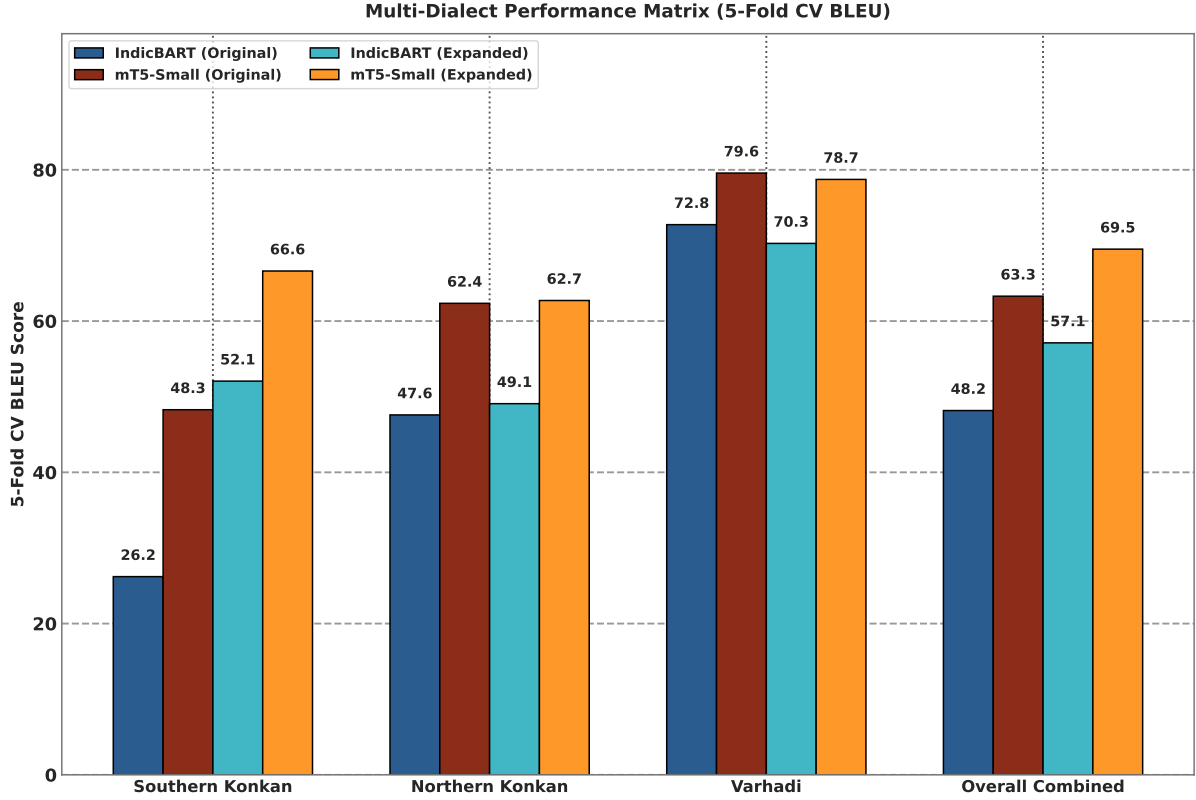


Figure 5: Multi-dialect 5-fold cross-validation benchmark comparison across model architectures and corpora partitions.

4.2 Comprehensive Benchmark on the Held-Out Test Set

While Section 4.1 reports 5-fold cross-validation results across training partitions, the evaluations in this section and subsequent subsections are conducted strictly on the official held-out test benchmark (IISc_RESPIN_test_mr, 2,170 unseen utterances) after training on the complete datasets. No cross-validation was performed on the held-out test set.

Single-dialect training shows that synthetic data augmentation substantially reduces IndicBART performance across all three dialects, while mT5-small remains comparatively stable and shows slight improvements on Northern Konkani and Varhadi. Multi-dialect training produces substantial gains for IndicBART on Southern Konkani (+26.0 BLEU) and Northern Konkani (+6.5 BLEU), while Varhadi decreases slightly (−3.0 BLEU). mT5-small shows smaller but consistently positive gains across all three dialects: +3.0, +0.7, and +1.2 BLEU for Southern Konkani, Northern Konkani, and Varhadi, respectively. The full performance evaluation matrix on the held-out test set is detailed in Table 7 and Figure 6.

Table 7: Comprehensive Evaluation Matrix on *IISc_RESPIN_test_mr* Benchmark Set (Single Test Run, No CV)

Model Scope	Training Data	Dialect / Subset	IndicBART (244M)		mT5-Small (300M)	
			BLEU	WER	BLEU	WER
Single	Original	Southern Konkani	57.80	26.60%	43.86	34.37%
		Northern Konkani	90.13	6.46%	79.46	11.95%
		Varhadi	83.59	10.19%	74.81	14.73%
	Expanded	Southern Konkani	24.06	60.32%	44.36	32.75%
		Northern Konkani	65.41	28.70%	79.76	12.61%
		Varhadi	67.97	25.43%	76.90	14.19%
Multi	Original	Southern Konkani	26.08	52.98%	40.94	35.34%
		Northern Konkani	47.92	32.64%	77.50	14.65%
		Varhadi	73.04	26.96%	73.47	16.16%
	Expanded	Southern Konkani	52.15	36.50%	43.88	34.96%
		Northern Konkani	54.35	25.65%	78.16	13.55%
		Varhadi	69.96	21.04%	74.74	15.57%

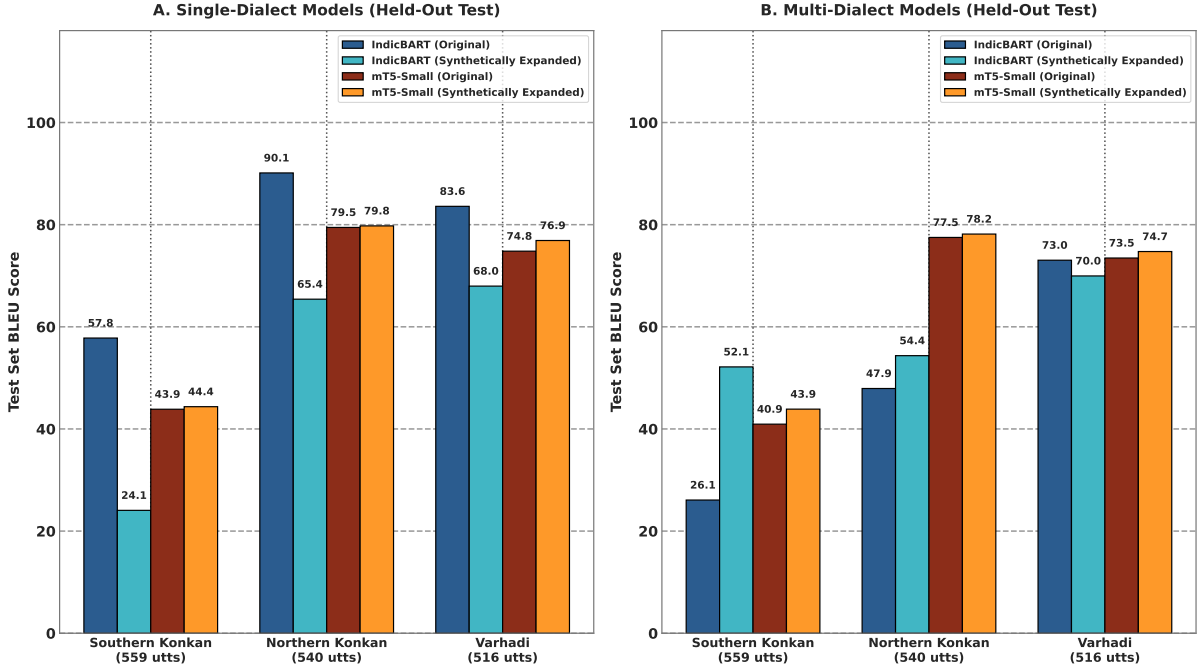


Figure 6: Comprehensive sequence-to-sequence evaluation matrix on the held-out *RESPIN* test benchmark (2,170 utterances), comparing *IndicBART* (244M) and *mT5-small* (300M) across single-dialect and multi-dialect training configurations.

4.3 Impact of Synthetic Data Augmentation on Test Set

Table 8 and Figure 7 evaluate the net BLEU impact (Δ BLEU) resulting from synthetic data augmentation directly on the held-out test set under multi-dialect training regimes (single test run, no cross-validation).

Overall, multi-dialect models exhibit substantial gains from synthetic data augmentation for most dialects. *IndicBART* shows particularly large improvements for Southern Konkani (+26.0

BLEU) and Northern Konkani (+6.5 BLEU), although its Varhadi performance decreases by 3.0 BLEU. mT5-small shows smaller but consistently positive gains across all three dialects: +3.0, +0.7, and +1.2 BLEU for Southern Konkani, Northern Konkani, and Varhadi, respectively. Visual representation of net BLEU score changes is provided in Figure 7.

Table 8: Impact of Synthetic Data Augmentation on Held-Out Test Set: Net BLEU Change (Single Test Run, No CV)

Evaluated Subset	IndicBART BLEU			mT5-Small BLEU		
	Original	Expanded	Δ	Original	Expanded	Δ
Southern Konkani (D1)	26.08	52.15	+26.07	40.94	43.88	+2.94
Northern Konkani (D2)	47.92	54.35	+6.43	77.50	78.16	+0.66
Varhadi (D4)	73.04	69.96	-3.08	73.47	74.74	+1.27

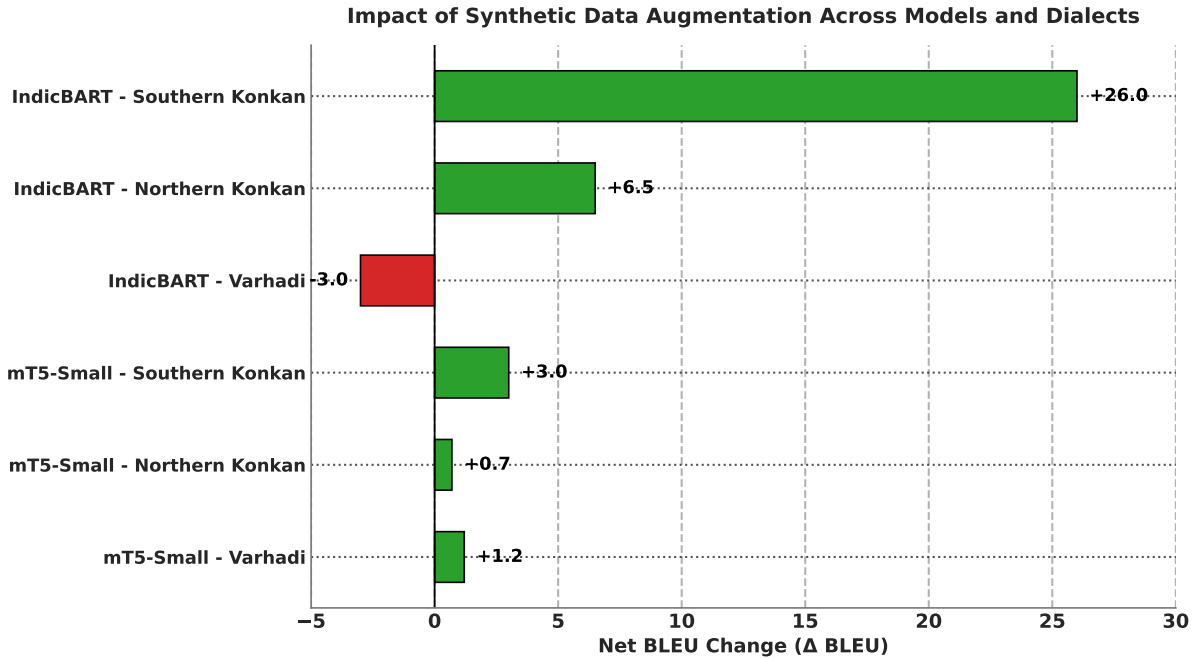


Figure 7: Impact of synthetic data augmentation (Δ BLEU score on the held-out test set under multi-dialect training regimes).

4.4 Verification Engine Impact: Raw Unverified vs. Filtered Clean Data

We analyzed the performance of our verification and filtering engine on the test benchmark (single test run, without cross-validation) by training on the Unverified dataset (19,914 pairs containing 3,751 contaminated pairs) versus the Filtered Clean dataset (32,335 pairs). Table 9 and Figure 8 demonstrate the BLEU and chrF++ score recovery provided by multi-tier verification.

Table 9: Impact of Multi-Tier Verification Engine: Raw Unverified vs. Filtered Clean Synthetic Data (Single Test Run, No CV)

Model	Pipeline State	Pairs	BLEU	chrF++	Δ BLEU	Δ chrF++
IndicBART (244M)	Raw Unverified Data	19,914	43.09	64.54	−14.03	−10.95
	Filtered Clean Data	32,335	57.12	75.49	+14.03	+10.95
mT5-Small (300M)	Raw Unverified Data	19,914	61.21	80.18	−8.30	−4.40
	Filtered Clean Data	32,335	69.51	84.58	+8.30	+4.40

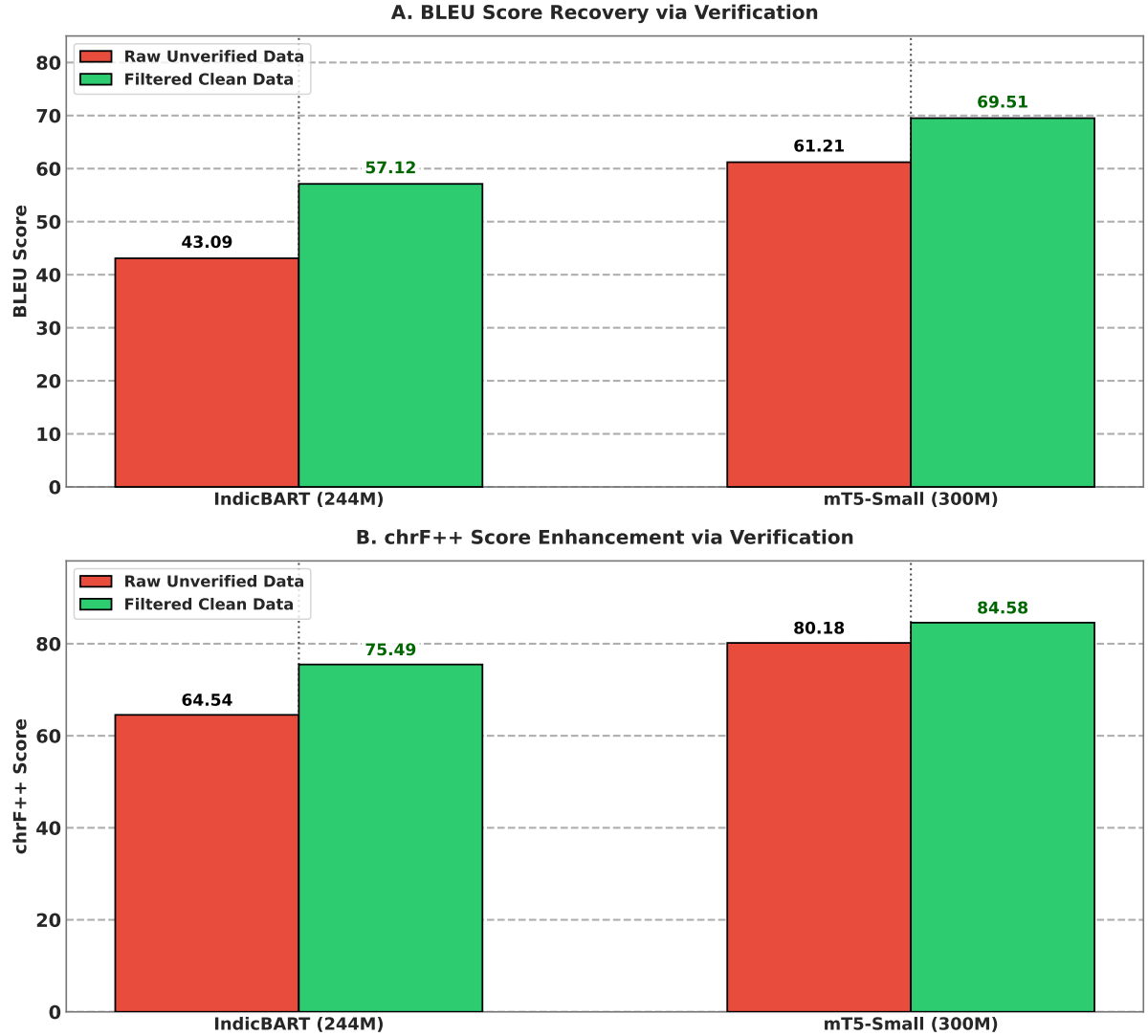


Figure 8: Multi-tier verification engine ablation: (A) BLEU score recovery and (B) chrF++ score enhancement comparing models trained on raw unverified synthetic outputs vs. filtered clean synthetic data.

Training on raw unverified synthetic data causes severe performance degradation (−14.03 BLEU for IndicBART, −8.30 BLEU for mT5-small). Unfiltered LLM outputs had hallucinated sentences which corrupted the gradients during backpropagation. Multi-tier verification (rule-based script filtering and LLM semantic auditing) removes 3,751 flawed pairs successfully recovering BLEU scores to 57.12 (IndicBART) and 69.51 (mT5-small).

IndicBART shows sensitivity to noise, suffering a 24.6% relative BLEU drop on unverified data (57.12 → 43.09). In contrast, mT5-small demonstrates robustness with a smaller relative

drop of 11.9% (69.51 $\rightarrow$ 61.21). This could be directly attributed to mT5’s larger 300M parameter capacity and extensive pre-training corpus providing stronger immunity against the noise. This highlights the importance of the quality audit of data while using LLMs for generating or correcting the data.

5 Conclusion

This study presents a scalable, reproducible framework for multi-dialect Marathi text normalization, addressing the critical scarcity of parallel corpora for low-resource Indic varieties. By replacing expensive manual annotations with a synthetic parallel generation pipeline combining Gemma-4 translation with a multi-tier verification engine (rule-based script filtering and LLaMA-3.1-8B semantic auditing), we successfully expanded the clean parallel corpus from 16,163 to 32,335 verified pairs across Southern Konkan, Northern Konkan, and Varhadi dialects. Empirical evaluations demonstrate that mT5-small achieves 69.51 BLEU and 84.58 chrF++ on the expanded multi-dialect benchmark, outperforming IndicBART by 12.39 BLEU. Crucially, multi-tier verification prevented synthetic quality collapse (+14.03 BLEU recovery over unverified data), while dataset expansion mitigated cross-dialect parameter competition and maintained near-perfect Standard Marathi normalization (97.39 BLEU, 1.37% WER) on held-out RESPIN benchmarks. While these findings establish a strong blueprint for Indic sub-dialect normalization, minor constraints remain: LLM-driven verification pipelines introduce computational and latency overheads, synthetic expansion can occasionally yield slight precision dilution on already high-performing dialects like Varhadi, and intrinsic text metrics should be complemented by downstream task evaluations (e.g., ASR post-processing) in future work. Overall, our verified synthetic pipeline provides a scalable foundation for expanding digital linguistic inclusion across low-resource regional dialects.

References

- Abdul-Mageed, M., Zhang, C., Elmadany, A., and Nagoudi, E. M. (2023). Nadi 2023: The fourth nuanced arabic dialect identification and translation shared task. In *Proceedings of the Arabic NLP Conference (ArabicNLP @ EMNLP 2023)*, pages 512–528.
- Abe, K., Inui, K., and Matsumoto, Y. (2018). Multi-dialect neural machine translation for japanese dialects. In *Proceedings of the 27th International Conference on Computational Linguistics (COLING 2018)*, pages 2105–2116.
- Ansary, N., Adib, Q. A. R., Reasat, T., Sushmit, A. S., Humayun, A. I., Mehnaz, S., Fatema, K., Or Rashid, M. M., and Sadeque, F. (2024). Unicode normalization and grapheme parsing of indic languages. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 17019–17030. ELRA Language Resource Association.
- Blaschke, V., Purschke, C., Schütze, H., and Plank, B. (2024). What do dialect speakers want? a survey of attitudes towards language technology for german dialects. *arXiv preprint arXiv:2402.11968*.
- Bouamor, H., Habash, N., Salameh, M., et al. (2018). The madar parallel corpus of arabic dialects. In *Proceedings of the 11th Language Resources and Evaluation Conference (LREC 2018)*, pages 3375–3381.
- Dabre, R., Shrotiya, H., Kunchukuttan, A., Puduppully, R., Khapra, M. M., and Kumar, P. (2022). Indicbart: A pre-trained model for indic natural language generation. In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 1849–1863.
- de Gibert, O. et al. (2024). Scaling low-resource mt via synthetic data generation with llms. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*.
- Dhasmana, A., Srivastava, A., and Chiang, D. (2026). Dialect matters: Cross-lingual asr transfer for low-resource indic language varieties. In *Proceedings of the Thirteenth Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ ACL 2026)*, pages 145–156. Association for Computational Linguistics.
- Dimakis, A., Pavlopoulos, J., and Anastasopoulos, A. (2025). Dialect normalization using large language models and morphological rules. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 23696–23714. Association for Computational Linguistics.
- Dubey, A., Jauhri, A., Pandey, A., Kadian, A., Al-Dahle, A., Letman, A., Mathur, A., Schelten, A., Yang, A., Fan, A., and et al. (2024). The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

- Eryani, F., Habash, N., Al-Shargi, F., et al. (2020). A normalization corpus for five arabic dialects. In *Proceedings of the 12th Language Resources and Evaluation Conference (LREC 2020)*, pages 5750–5757.
- Gaikwad, G., Panchal, D., Deshmukh, O., and Patil, S. (2025). Marathi dialect to pure marathi using rag & ai. *International Research Journal of Engineering and Technology (IRJET)*, 12(11):787–791.
- Google DeepMind Gemma Team (2026). Gemma 4: Open language models. *arXiv preprint arXiv:2603.00118*.
- Habash, N., Diab, M., and Rambow, O. (2012). Conventional orthography for dialectal arabic. In *Proceedings of the Eighth International Conference on Language Resources and Evaluation (LREC '12)*, pages 711–718.
- Hämäläinen, M., Partanen, N., and Alnajjar, K. (2020). Normalizing swedish dialects spoken in finland. In *Proceedings of the 7th Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ COLING 2020)*, pages 55–62.
- Hämäläinen, M., Partanen, N., and Rueter, J. (2022). Generating dialectal finnish for dialect normalization of estonian dialects. In *Proceedings of the 9th Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ COLING 2022)*, pages 71–79.
- Hsieh, C.-Y. et al. (2023). Distilling step-by-step! outperforming larger language models with less training data and smaller model sizes. *arXiv preprint arXiv:2305.02301*.
- Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y. J., Madotto, A., and Fung, P. (2023). Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):1–38.
- Joshi, A., Dabre, R., Kanojia, D., Li, Z., Zhan, H., Haffari, G., and Dippold, D. (2025). Natural language processing for dialects of a language: A survey. *ACM Computing Surveys*, 57(6):1–37.
- Joshi, P., Santy, S., Budhiraja, A., Bali, K., and Choudhury, M. (2020). The state and fate of linguistic diversity and inclusion in the NLP world. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 6282–6293.
- Joshi, R., Goel, P., and Joshi, R. (2022). L3cube-mahanlp: Marathi natural language processing datasets, models, and library. In *Proceedings of the WILDRE-6 Workshop, LREC 2022*, pages 10–17.
- Kakwani, D., Kunchukuttan, A., Golla, S., Gokul, N. C., Bhattacharyya, P., Khapra, M. M., and Kumar, P. (2020). Indicnlp suite: Monolingual corpora, evaluation benchmarks and pre-trained multilingual language models for indic languages. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 4948–4961. Association for Computational Linguistics.
- Koenecke, A., Nam, A., Lake, E., Nudell, J., Quartey, M., Mengesha, Z., Toups, C., Rickford, J. R., Jurafsky, D., and Goel, S. (2020). Racial disparities in automated speech recognition. *Proceedings of the National Academy of Sciences*, 117(14):7684–7689.
- Kresic, M. and Abbas, N. (2024). Normalizing swiss german dialects with the power of large language models. *Procedia Computer Science*, 244:287–295.
- Kulkarni-Joshi, S. (2023). Variation and change in dialects of marathi: A social-dialectological approach. In Chandra, P., editor, *Variation in South Asian Languages: From Macro to Micro-Differences*, pages 207–236. Springer.
- Kumar, A., Shrivastava, M., and Bhattacharyya, P. (2024). Indic-dialect: Benchmark dataset for indic dialect classification and translation. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 15201–15212. ELRA Language Resource Association.
- Kumar, S., Singh, A., G. D., Amartyaveer, Bandekar, J., Murthy, S., Sharma, S., Badiger, S., Udupa, S., Nagireddi, A., Raghavan, S. K. M., Saxena, R., Nanavati, J., Nanavati, R., Sridharan, J., Mehta, A., Khuraishi, A. K. S., Mora, S. P. R., Venkataramakrishnan, P., Date, G., P. K., and Ghosh, P. K. (2025). RESPIN-S1.0: A read speech corpus of 10000+ hours in dialects of nine indian languages. In *Advances in Neural Information Processing Systems (NeurIPS 2025) Track on Datasets and Benchmarks*, pages 1–13.
- Kuparinen, O., Milić, A., and Scherrer, Y. (2023). Dialect-to-standard normalization: A large-scale multilingual evaluation. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 13814–13828. Association for Computational Linguistics.
- Lusetti, M., Ruzsics, T., Samardžić, T., and Scherrer, Y. (2018). Impact of target language models on swiss german dialect normalization. In *Proceedings of the 5th Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ COLING 2018)*, pages 133–141.
- Mukherjee, S., Mitra, A., Jawahar, G., Agarwal, S., Palangi, H., and Awadallah, A. (2023). Orca: Progressive learning from complex explanation traces of gpt-4.
- Mulla, R. and Kumar, B. S. (2022). Text-independent automatic dialect recognition of marathi language using spectro-temporal characteristics of voice. *International Journal on Recent and Innovation Trends in Computing and Communication (IJRITCC)*, 10(2s):313–321.

- Partanen, N., Härmäläinen, M., and Alnajjar, K. (2019). Dialect text normalization to normative standard finnish. In *Proceedings of the 5th Workshop on Noisy User-generated Text (W-NUT @ EMNLP 2019)*, pages 141–146. Association for Computational Linguistics.
- Pawar, P. S., Saini, J. R., and Vaidya, S. (2026). Automatic marathi dialect classification in noisy and non-noisy environments. *Procedia Computer Science*, 282:288–303.
- Pecher, B. et al. (2024). Better as generators than classifiers: Leveraging llms and synthetic data for low-resource multilingual classification. In *Findings of the Association for Computational Linguistics: EACL 2024*.
- Post, M. (2018). A call for clarity in reporting bleu scores. In *Proceedings of WMT 2018*, pages 186–191.
- Pratap, V., Saraf, A., and Kumar, P. (2024). Wfst-guided neural text normalization for indic languages. In *Proceedings of Interspeech 2024*, pages 3120–3124.
- Scherrer, Y. and Ljubešić, N. (2016). Normalising swiss german dialect text using character-level machine translation. In *Proceedings of the 3rd Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ COLING 2016)*, pages 117–126.
- Taori, R., Gulrajani, I., Zhang, T., Dubois, Y., Li, X., Guestrin, C., Liang, P., and Hashimoto, T. B. (2023). Alpaca: A strong, replicable instruction-following model. *Stanford Center for Research on Foundation Models*.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. (2017). Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS 2017)*, pages 5998–6008.
- Vyavhare, V. A., Divekar, S. N., Pawar, H., Kale, K., and Jagtap, S. (2026). Marathi dialect translator. *International Journal of Scientific Research in Science, Engineering and Technology (IJSRSET)*, 13(3):254–261.
- Wang, Y., Kordi, Y., Mishra, S., Liu, A., Smith, N. A., Khashabi, D., and Hajishirzi, H. (2022). Self-instruct: Aligning language models with self-generated instructions.
- Xue, L., Constant, N., Roberts, A., Kale, M., Al-Rfou, R., Siddhant, A., Barua, A., and Raffel, C. (2021). mt5: A massively multilingual pre-trained text-to-text transformer. In *Proceedings of NAACL-HLT 2021*, pages 483–498.
- Zampieri, M., Nakov, P., and Scherrer, Y. (2020). Natural language processing for similar languages, varieties, and dialects: A survey. *Natural Language Engineering*, 26(6):605–637.
- Zhao, Y. and Chodroff, E. (2022). Acoustic-phonetic variation in mandarin dialect corpora. *Journal of Phonetics*, 94:101172.